

DAY 1: DIFFERENTIATION - PROBLEMS

RATES OF CHANGE IN MECHANICS

Instructions

Solve all problems showing complete work. For physics problems, include units in your answers.

1 PART 1: FOUNDATIONS & ALL RULES (The Math)

1.1 Problem 1.1: The Limit Definition & Power Rule

The derivative of a function $f(x)$ at a point x is defined as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

1. Using this definition, prove that if $f(x) = x^3$, then $f'(x) = 3x^2$. Show every algebraic step.
2. State the power rule: $\frac{d}{dx}[x^n] = ?$
3. Differentiate: $f(x) = 5x^4 - 3x^2 + 7x - 10 + \frac{4}{x^3} + 2\sqrt{x}$.
Hint: Rewrite $\frac{4}{x^3} = 4x^{-3}$ and $\sqrt{x} = x^{1/2}$.
4. Differentiate: $g(x) = \frac{1}{3}x^6 + 4x^{-3} - 2x^{1/2} + 5x^{-2} + 7$.

1.2 Problem 1.2: Product and Quotient Rules

1. State the product rule: $\frac{d}{dx}[u(x)v(x)] = ?$
2. State the quotient rule: $\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = ?$

3. Differentiate: $f(x) = (3x^2)(5x^3 - 2x)$.

Do it with the product rule, then simplify first and use power rule to check your answer.

4. Differentiate: $g(x) = \frac{x^2+1}{x^3-4x}$.

5. Differentiate: $h(x) = \frac{x\sqrt{x}}{x^2-1}$.

Hint: $x\sqrt{x} = x \cdot x^{1/2} = x^{3/2}$

1.3 Problem 1.3: The Chain Rule (The Most Important Rule)

1. State the chain rule in two forms:

- Using function notation: $\frac{d}{dx}[f(g(x))] = ?$
- Using Leibniz notation: $\frac{dy}{dx} = ?$

2. Differentiate: $f(x) = (3x^2 - 5x)^4$.

3. Differentiate: $g(x) = \sqrt{x^3 - 1}$.

4. Differentiate: $h(x) = \frac{1}{\sqrt{4-x^2}}$.

5. Differentiate: $y = \sin(3x^2 + 1)$.

Recall: $\frac{d}{dx}[\sin(u)] = \cos(u) \frac{du}{dx}$.

1.4 Problem 1.4: Implicit Differentiation

1. Explain: When do we use implicit differentiation?
2. Find $\frac{dy}{dx}$ for the circle: $x^2 + y^2 = 25$.
3. Find the slope of the tangent line to this circle at the point $(3, -4)$.
4. Find $\frac{dy}{dx}$ for: $x^3 + y^3 = 6xy$. (This is called the folium of Descartes).
5. Find $\frac{dy}{dx}$ for: $\sin(xy) = x^2 - y$.
6. Find the second derivative $\frac{d^2y}{dx^2}$ implicitly for $x^2 + y^2 = 25$.

1.5 Problem 1.5: Parametric & Logarithmic Differentiation

1. Parametric Differentiation:

If $x = f(t)$ and $y = g(t)$, then $\frac{dy}{dx} = ?$

Given $x = t^2 + 1$, $y = t^3 - t$, find $\frac{dy}{dx}$ at $t = 2$.

Find the equation of the tangent line at this point.

2. Logarithmic Differentiation:

Differentiate $y = \frac{(x^2+1)^3 \cdot e^x \cdot \sin^2(x)}{(x^3-1)^2}$ by taking $\ln$ of both sides first, then differentiating implicitly.

2 PART 2: PHYSICS APPLICATIONS - KINEMATICS

2.1 Problem 1.6: From Position to Velocity to Acceleration

The position of a particle is given by $x(t) = 2t^3 - 15t^2 + 24t + 10$ meters, where t is in seconds.

1. Find the velocity function $v(t) = \frac{dx}{dt}$.
2. Find the acceleration function $a(t) = \frac{dv}{dt}$.
3. When is the particle at rest? (Solve $v(t) = 0$).
4. When is the particle moving in the positive direction? ($v(t) > 0$)?
5. When is the acceleration zero? What is the velocity at this time?
6. **Jerk:** Find the jerk $j(t) = \frac{da}{dt}$. When is jerk zero?

2.2 Problem 1.7: Non-Constant Acceleration (Differential Equation Intro)

An object's acceleration is given by $a(v) = -kv^2$, where k is a constant. This is a common model for air resistance at high speeds.

1. We know $a = \frac{dv}{dt}$. So we have $\frac{dv}{dt} = -kv^2$. Separate the variables to get $v^{-2}dv = -kdt$.
2. Integrate both sides: $\int v^{-2}dv = \int -kdt$.
3. If at $t = 0$, $v = v_0$, find the constant C and derive the final formula for $v(t)$.
What happens as $t \rightarrow \infty$?

2.3 Problem 1.8: Related Rates (Implicit Diff in Time)

A ladder 10 meters long rests against a vertical wall. The bottom of the ladder slides away from the wall at a rate of 1 m/s. How fast is the top of the ladder sliding down the wall when the bottom is 6 meters from the wall?

1. Let x be the distance from the wall to the bottom, and y be the height of the top on the wall. Write an equation relating x and y .
2. Differentiate this equation implicitly with respect to time t .

3. We are given $\frac{dx}{dt} = 1$ m/s. Find $\frac{dy}{dt}$ when $x = 6$.

2.4 Problem 1.9: Parametric Motion in 2D

A projectile is launched. Its position is given by:

$$\begin{aligned}x(t) &= (20 \text{ m/s}) \cdot t \\y(t) &= (30 \text{ m/s}) \cdot t - \frac{1}{2}(9.8 \text{ m/s}^2)t^2\end{aligned}$$

1. Find the velocity components $v_x(t) = \frac{dx}{dt}$ and $v_y(t) = \frac{dy}{dt}$.
2. Find the speed of the projectile at any time t .
3. Find the direction of the velocity (the angle θ relative to the horizontal) as a function of t .
4. Find the acceleration components $a_x(t)$ and $a_y(t)$.
5. Find the tangential acceleration $a_t = \frac{d|\vec{v}|}{dt}$ (rate of change of speed).

3 PART 3: OPTIMIZATION - FINDING MAXIMA AND MINIMA IN PHYSICS

3.1 Problem 1.10: Maximum Height of a Projectile

A projectile is launched vertically upward from ground level with an initial velocity of 49 m/s. Its height at time t is given by:

$$h(t) = 49t - 4.9t^2 \text{ meters}$$

1. Find the velocity function $v(t) = \frac{dh}{dt}$.
2. At what time does the projectile reach its maximum height? (Hint: At maximum height, the vertical velocity is zero).
3. What is the maximum height?
4. Verify this is a maximum using the second derivative test.
5. How long does it take for the projectile to return to the ground? What is its impact velocity?

3.2 Problem 1.11: Maximum Range of a Projectile

A projectile is launched from ground level with initial speed v_0 at an angle θ above the horizontal. The range (horizontal distance traveled before landing) is given by:

$$R(\theta) = \frac{v_0^2 \sin(2\theta)}{g}$$

where $g = 9.8 \text{ m/s}^2$ and $v_0 = 20 \text{ m/s}$.

1. Write $R(\theta)$ explicitly for $v_0 = 20 \text{ m/s}$ and $g = 9.8 \text{ m/s}^2$.
2. Find $\frac{dR}{d\theta}$.
3. Find the critical points by setting $\frac{dR}{d\theta} = 0$ for $0 < \theta < \frac{\pi}{2}$.
4. Verify that this gives a maximum using the second derivative test.
5. What is the maximum range?
6. What happens to the range if $\theta = 30^\circ$? $\theta = 60^\circ$? Compare.

3.3 Problem 1.12: Minimum Surface Area of a Cylinder

A cylindrical can must hold a volume of $V = 1000 \text{ cm}^3$ (1 liter). Find the dimensions (radius r and height h) that minimize the surface area (and thus the amount of material used).

Recall: Volume of cylinder: $V = \pi r^2 h$, Surface area (including top and bottom): $S = 2\pi r^2 + 2\pi r h$

1. Use the volume constraint to express h in terms of r .
2. Write the surface area S as a function of r only.
3. Find $\frac{dS}{dr}$.
4. Find the critical point by setting $\frac{dS}{dr} = 0$.
5. Find the corresponding height h .
6. Verify that this gives a minimum using the second derivative test.

3.4 Problem 1.13: Fermat's Principle - Minimizing Time

Fermat's principle states that light travels between two points along the path that minimizes travel time. This leads to Snell's Law of refraction.

A lifeguard at point A on the beach (10 m from the water) sees a swimmer in distress at point B in the water (20 m from the shore). The lifeguard can run at speed $v_1 = 5 \text{ m/s}$ on sand and swim at speed $v_2 = 2 \text{ m/s}$ in water. The points are separated horizontally by 50 m along the shore.

1. Set up coordinates: Let the shoreline be the x-axis. Place A at $(0, 10)$ and B at $(50, -20)$. The lifeguard runs to a point $P = (x, 0)$ on the shore, then swims directly to B . Write the total time $T(x)$ as a function of x .
2. Find $\frac{dT}{dx}$.
3. Set $\frac{dT}{dx} = 0$ to find the optimal x . Show that this leads to $\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$, where θ_1 and θ_2 are the angles of incidence and refraction.
4. Solve numerically for x (use trial and error or a calculator).
5. What is the minimum time?

3.5 Problem 1.14: Maximum Power Transfer

In an electrical circuit with a battery of voltage E and internal resistance r , connected to a load resistance R , the power delivered to the load is:

$$P(R) = \frac{E^2 R}{(R + r)^2}$$

1. For a fixed E and r , find $\frac{dP}{dR}$.
2. Set $\frac{dP}{dR} = 0$ to find the value of R that maximizes the power $P(R)$.
3. Verify that this gives a maximum using the first or second derivative test.
4. What is the maximum power?

3.6 Problem 1.15: Minimum Time to Cross a River

A swimmer can swim at speed $v_s = 2$ m/s in still water. A river of width $w = 100$ m flows with current speed $v_c = 1$ m/s. The swimmer wants to reach a point directly opposite her starting point on the other bank.

1. If she swims at an angle θ upstream from the perpendicular, her velocity perpendicular to the banks is $v_s \cos \theta$ and her velocity parallel to the banks (relative to water) is $v_s \sin \theta$ upstream. The current carries her downstream at v_c . Write the time $T(\theta)$ to cross the river.
2. To reach the point directly opposite, her net downstream displacement must be zero. Write the condition for this and solve for θ .
3. What is the crossing time at this angle?
4. Now suppose she doesn't care where she lands downstream, but wants to minimize the crossing time. What angle should she choose? What is the minimum time? Where does she land?
5. Is it possible to reach the opposite point faster than the angle found in part (2)? Explain.